

Optimizing Deep Learning for Credit Default Prediction

A Comparative Study of Random Search and Bayesian Optimization

“Improving Accuracy, Efficiency, and Interpretability in Credit Risk Modeling”

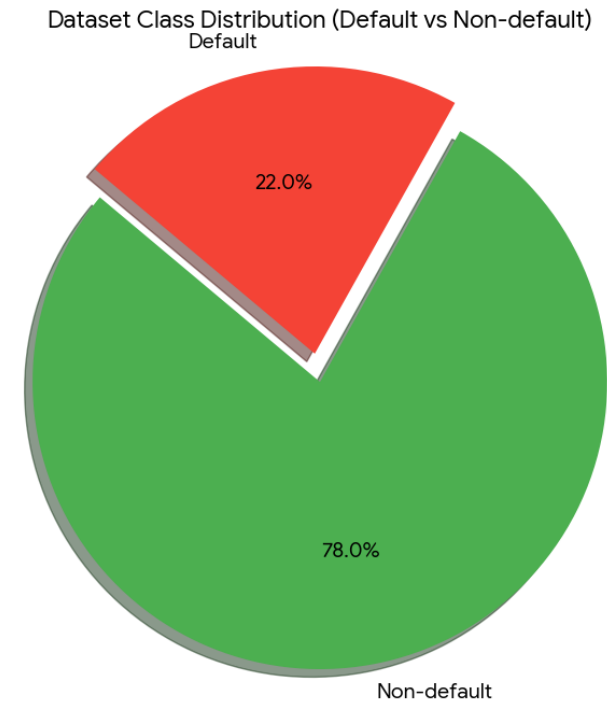
Machine Learning Optimization Strategies

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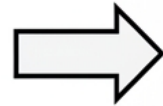
Project Objective & Dataset

- **Objective:** To explore how different hyperparameter optimization strategies affect the performance and efficiency of a PyTorch-based neural network.
- **Dataset:** UCI Credit Card Default dataset (30,000 records, 23 features).
- **Problem Statement:** Predicting credit default is critical for financial institutions to mitigate risk.
- **Key Preprocessing:** Scaling with StandardScaler to ensure stable convergence and stratified splitting (80/20).

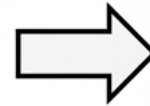
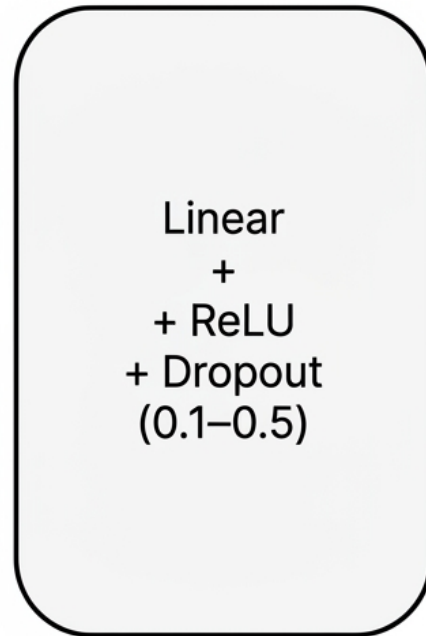


'CreditNet PyTorch Model'

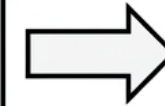
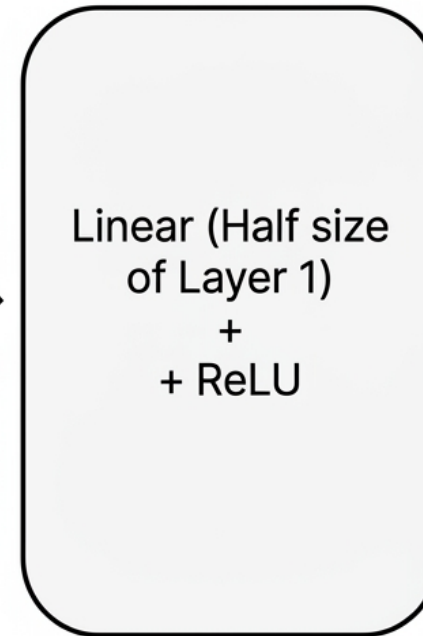
Input Layer: 23
nodes
(representing the
features).



Hidden Layer 1



Hidden Layer 2



Output Layer:
1 node with
Sigmoid
Sigmoid
activation.

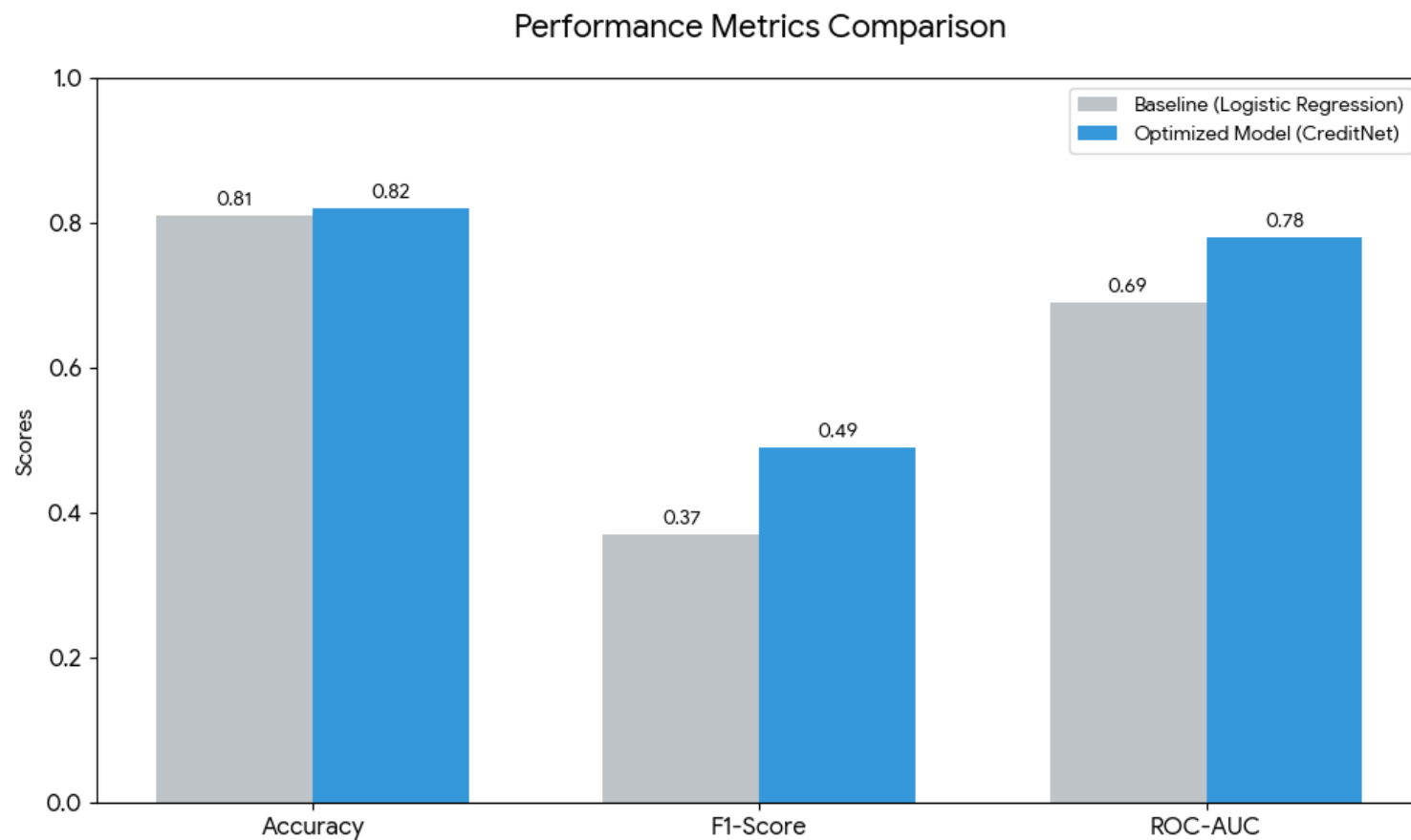


Prediction

Optimization Strategy Comparison

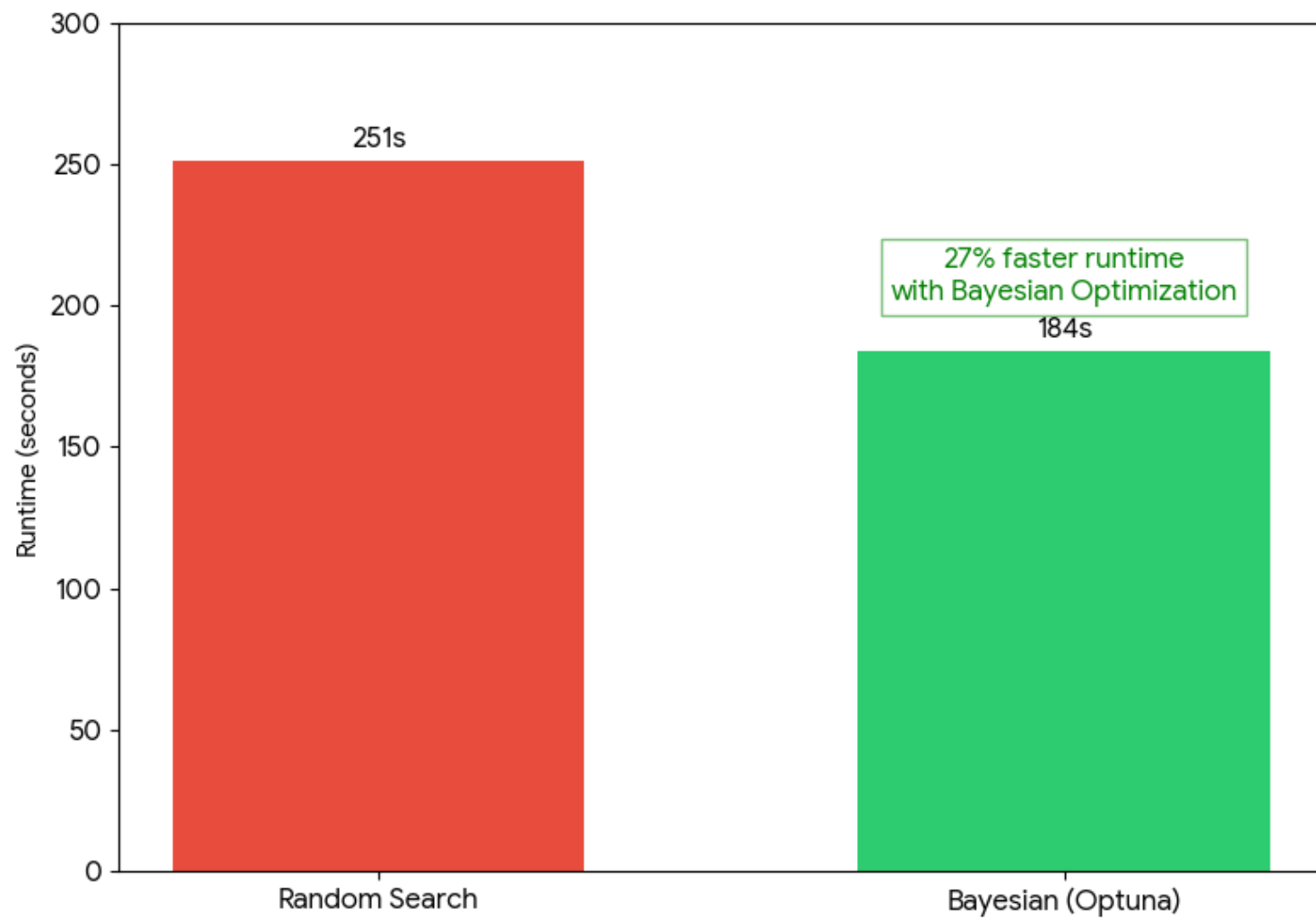
Feature	Random Search (Baseline)	Bayesian Optimization (Optuna)
Number of Trials	10 Trials	15 Trials
Search Logic	Random guesses in search space	Uses prior results to guide search
Efficiency	Less efficient (Stochastic)	Faster and smarter (Guided)
Search Pattern	Uniformly distributed dots	Clustered around promising areas

Performance Results



Performance Results

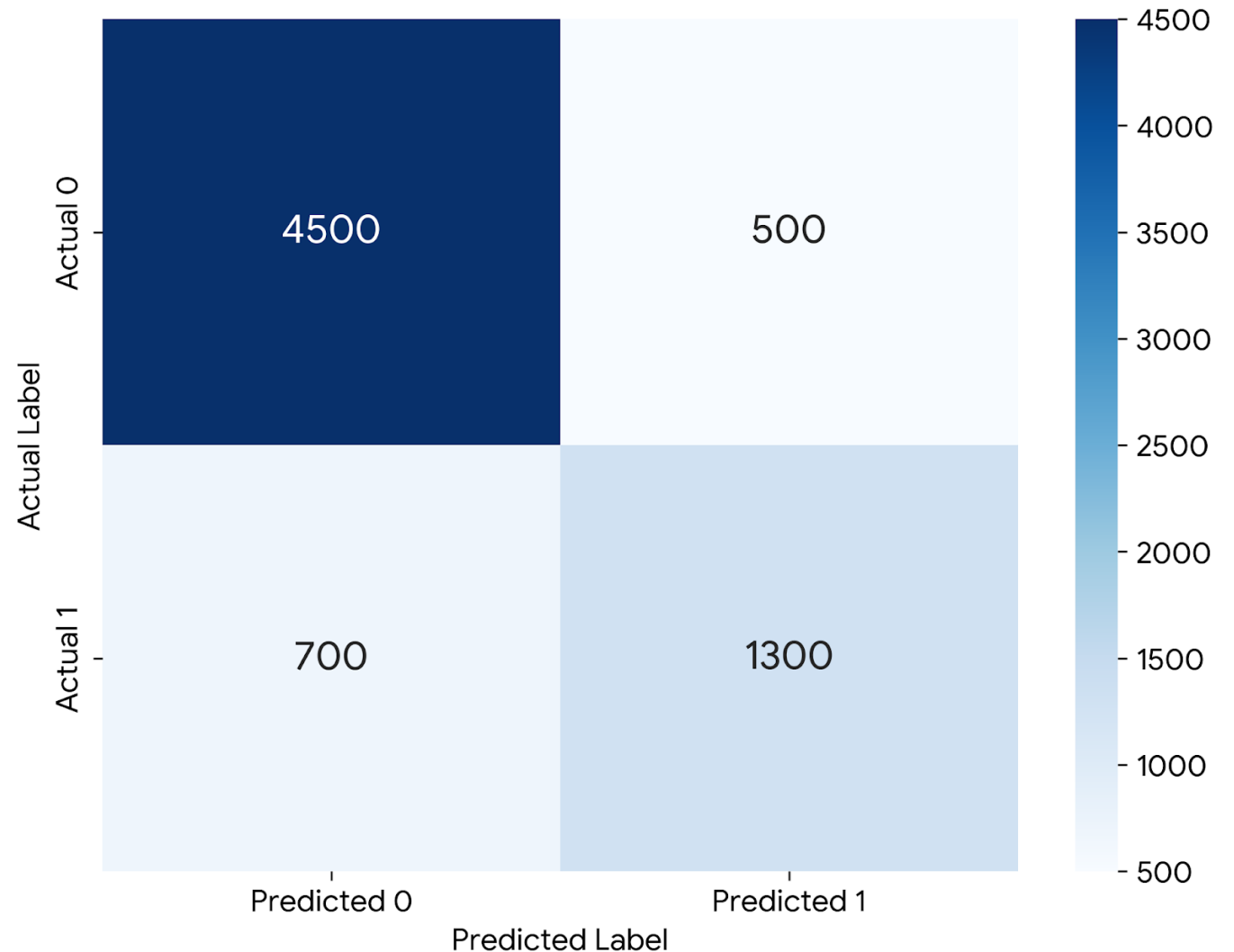
Optimization Runtime Comparison



Model Evaluation (Confusion Matrix)

- Baseline Comparison:** Logistic Regression achieved ~0.81 accuracy.
- Neural Network Strength:** Higher recall indicates the deep learning model is better at detecting actual defaulters compared to simpler linear models.
- Analysis:** The optimized model captures non-linear patterns that Logistic Regression misses.

Confusion Matrix: Optimized CreditNet Model

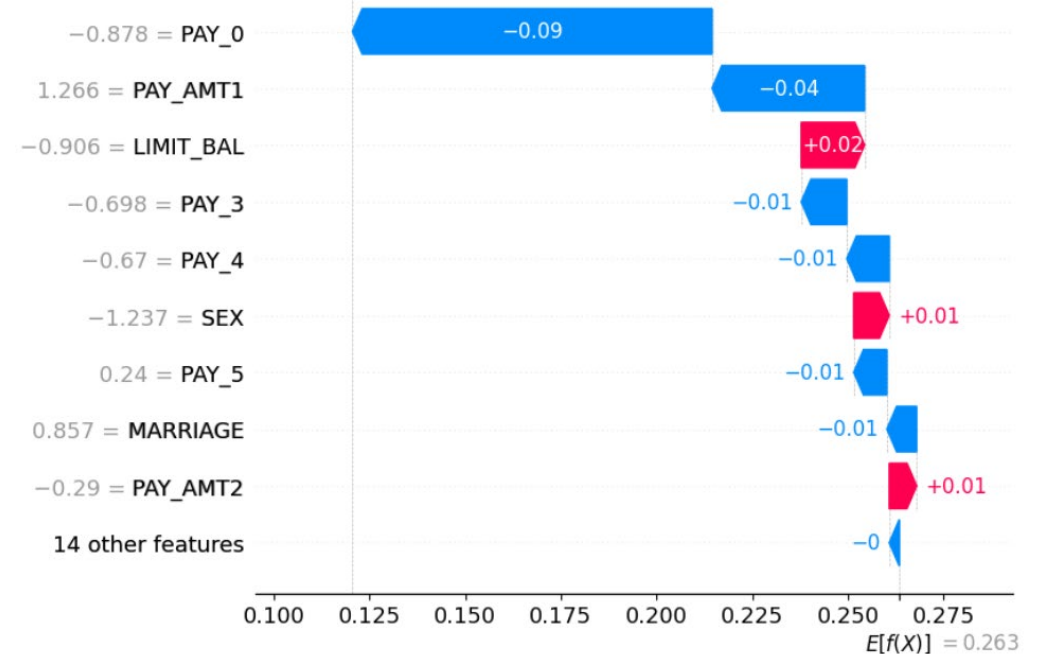


Explainable AI (XAI) with SHAP

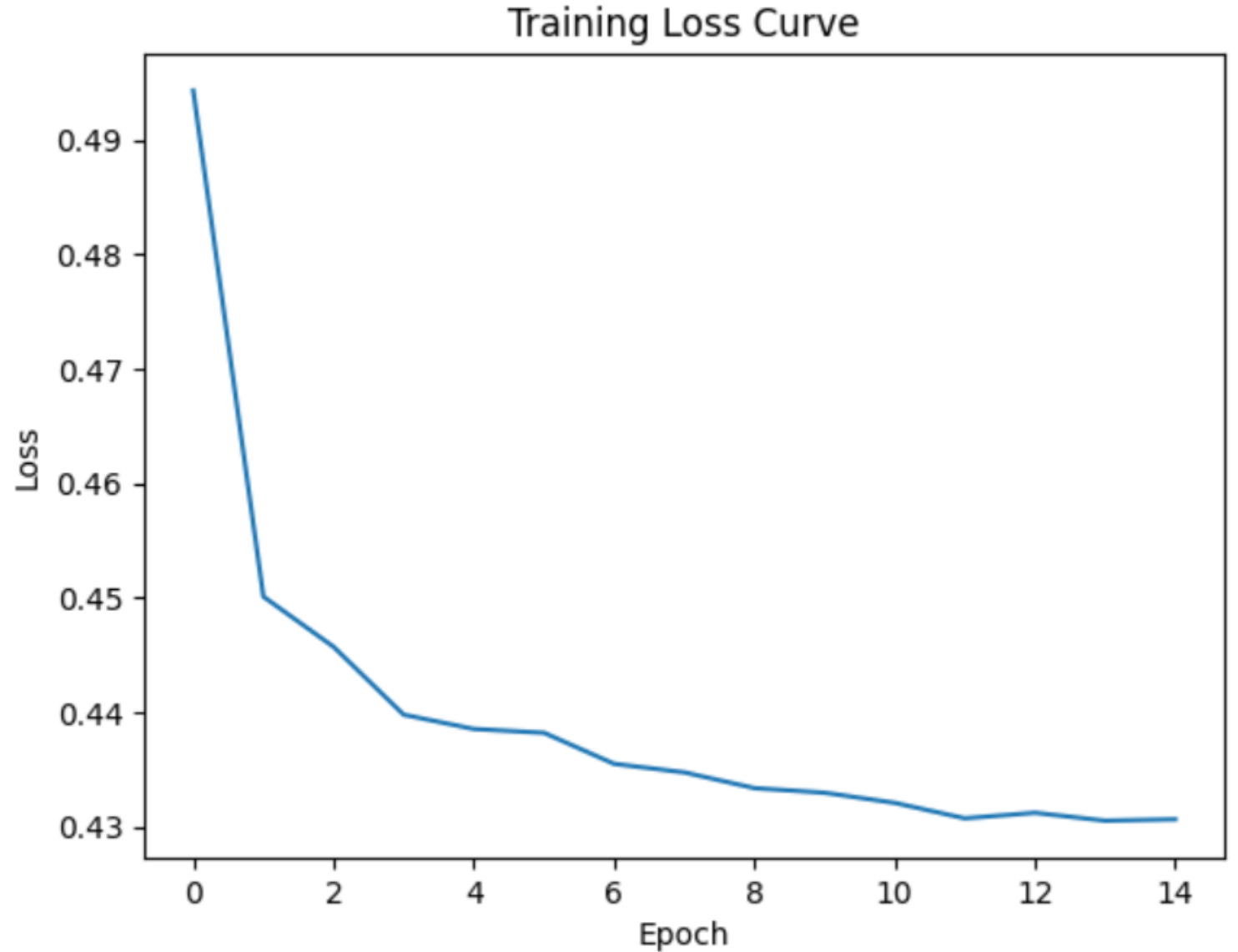
Global Feature Importance - Summary Plot



Local Explanation: Influence of Features on Single Prediction
 $f(x) = 0.121$



Training Behavior



Conclusion

- Bayesian Optimization:** Higher predictive power with significantly lower computational cost.
- Neural Networks:** Successfully capture complex financial behaviors missed by Logistic Regression.
- SHAP:** Crucial for building "black-box" trust in financial AI applications.